GNU Emacs Keyboard Shortcuts – Complete Editing Guide

This is a practical Emacs cheat sheet, from beginner cursor movement to advanced editing and code navigation.

Since you're using GNU Emacs 31.1 on Windows, these are the standard Emacs keybindings.

First: Understand Emacs key notation

Emacs uses a special notation for keyboard shortcuts.

Notation	Meaning
`C`	Ctrl
`M`	Alt (Meta)
`S`	Shift
`RET`	Enter
`SPC`	Space
`DEL`	Backspace
`ESC`	Escape

Examples:

- `C-f` = Ctrl + F
- `M-f` = Alt + F
- `C-x C-s` = Ctrl + X, then Ctrl + S
- `C-x k` = Ctrl + X, then K

Important: `C-x C-s` means press the first combination, release, then press the second. Don't hold all keys simultaneously.

On Windows, if Alt combinations don't work, press `ESC` and then the corresponding key. For example, `ESC f` is equivalent to `M-f`.

1. Basic cursor motion

These are the most important Emacs shortcuts. Learn these first.

Character and line movement

C-fForward one character

C-bBackward one character

C-nNext line

C-pPrevious line

Line movement

- `C-a` — Beginning of line
- `C-e` — End of line
- `M-m` — First non-whitespace character of line
- `M-r` — Move vertically to a chosen screen position

Word movement

- `M-f` — Forward one word
- `M-b` — Backward one word

Buffer movement

- `M-<` — Beginning of entire file/buffer
- `M->` — End of entire file/buffer
- `C-v` — Scroll down one screen
- `M-v` — Scroll up one screen
- `C-l` — Recenter screen around cursor

2. Jump around your code

These commands help you navigate large source files.

Shortcut	Action
`M-g g`	Jump to line number
`M-g M-g`	Jump to line number
`M-g c`	Jump to character position
`C-M-f`	Forward one expression/parenthesized block
`C-M-b`	Backward one expression
`C-M-a`	Beginning of function
`C-M-e`	End of function
`C-M-u`	Move up one syntactic level
`C-M-d`	Move down into an expression

For example, jump directly to line 150:

1. Press `M-g g`.
2. Type `150`.
3. Press `RET`.

Useful: `C-M-f` and `C-M-b` are especially helpful when editing Lisp, C, JavaScript, and other languages with nested expressions.

Note: Some structural commands depend on the programming mode.

3. Delete and edit text

Shortcut	Action
`DEL`	Delete character before cursor
`C-d`	Delete character after cursor
`M-DEL`	Delete previous word
`M-d`	Delete next word
`C-k`	Kill text to end of line
`C-S-DEL`	Kill entire line (in standard bindings)
`C-o`	Insert a newline without moving the cursor
`C-j`	Insert newline and indent
`C-t`	Transpose characters
`M-t`	Transpose words
`C-/`	Undo
`C-_`	Undo (alternative)
`M-u`	Uppercase next word
`M-l`	Lowercase next word
`M-c`	Capitalize next word

Important difference:

- Delete removes text without saving it in the kill ring.
- Kill removes text and allows you to paste it later.

Paste killed text with `C-y`.

4. Selecting text

Emacs uses something called the Mark and Region system.

- Point = current cursor position.
- Mark = selection starting position.
- Region = text between point and mark.

Shortcut	Action
`C-SPC`	Set mark
`C-g`	Cancel current command
`C-x C-x`	Exchange point and mark
`C-w`	Cut selected text
`M-w`	Copy selected text
`C-y`	Paste (yank)
`M-h`	Select paragraph
`C-x h`	Select entire buffer

Example: Select and copy a paragraph

1. Move to the beginning of the paragraph.

2. Press `M-h`.
3. Press `M-w` to copy.
4. Move somewhere else.
5. Press `C-y` to paste.

Select with keyboard

1. `C-SPC` — Start selection.
2. Move with `C-f`, `C-n`, `M-f`, etc.
3. `C-w` — Cut or `M-w` — Copy.

Tip: Emacs can also use Shift + arrow keys for selection when transient mark mode is enabled.

5. Search and replace

One of the best features of Emacs is incremental search.

Searching

Shortcut	Action
`C-s`	Search forward
`C-r`	Search backward
`C-s C-s`	Repeat previous search
`C-g`	Cancel search
`M-s o`	Search and display matching lines
`C-M-s`	Search using regular expressions
`C-M-r`	Backward regexp search

Search and replace

Shortcut	Action
`M-%`	Query replace
`C-M-%`	Query replace using regex
`M-x replace-string`	Replace all matching text
`M-x replace-regexp`	Replace using regex

Example: Replace `hello` with `world`

1. Press `M-%`.
2. Enter `hello`.
3. Press `RET`.
4. Enter `world`.
5. Press `RET`.

Emacs will ask what to do at each match:

Key	Action
`y`	Replace this occurrence
`n`	Skip this occurrence

Key	Action
`!`	Replace all remaining occurrences
`q`	Quit
`.`	Replace and quit
`^`	Back up to previous match

6. Working with multiple windows

In Emacs, a window is a section of your current frame that displays a buffer.

8 Reasons Why Emacs is the Best Text Editor for Programming | HackerNoon

[hackernoon.com](<https://hackernoon.com/8-reasons-why-emacs-is-the-best-text-editor-for-programming-0w4o37ld?utm_source=chatgpt.com>)

Shortcut	Action
`C-x 2`	Split horizontally (top/bottom)
`C-x 3`	Split vertically (left/right)
`C-x o`	Switch to next window
`C-x 0`	Close current window
`C-x 1`	Keep only current window
`C-x 4 f`	Open file in another window
`C-x 4 b`	Open buffer in another window
`C-x ^`	Increase window height
`C-x }`	Increase window width
`C-x {`	Decrease window width
`C-x +`	Balance window sizes

Example: Edit two files side by side

1. Open your first file using `C-x C-f`.
2. Press `C-x 3`.
3. Press `C-x 4 f`.
4. Enter the second filename.
5. Switch between them using `C-x o`.

7. Working with buffers and files

Buffers

A buffer is Emacs' in-memory representation of text. A file and a buffer are not exactly the same thing.

Shortcut	Action
`C-x b`	Switch buffer
`C-x C-b`	List all buffers
`C-x k`	Kill current buffer
`C-x LEFT`	Previous buffer

Shortcut	Action
C-x RIGHT	Next buffer
C-x 4 b	Switch buffer in another window

Files

Shortcut	Action
C-x C-f	Open or create file
C-x C-s	Save file
C-x C-w	Save as
C-x s	Save modified buffers
C-x C-r	Open file read-only
M-x revert-buffer	Reload file from disk
C-x i	Insert another file at cursor
C-x C-q	Toggle read-only mode

Exiting Emacs

- C-x C-c — Exit Emacs.
- C-z — Suspend Emacs (behavior depends on the environment).

8. Advanced editing

Now we're getting into the powerful features that make Emacs different from ordinary text editors.

A. Advanced code editing

Shortcut	Action
TAB	Indent current line (mode-dependent)
C-M-\	Indent selected region
M-;	Insert or align comment
C-M-o	Split line at point
M-q	Reformat paragraph
C-M-f	Move forward across expression
C-M-b	Move backward across expression
C-M-k	Kill expression
C-M-t	Transpose expressions

B. Keyboard macros

Macros let you record a sequence of editing actions and replay them.

Shortcut	Action
F3	Start recording macro
F4	Stop recording / execute macro
C-x e	Execute last keyboard macro

Shortcut	Action
`C-u 10 C-x e`	Execute macro 10 times
`M-x name-last-kbd-macro`	Give macro a name

Example:

1. Press `F3`.
2. Perform some editing actions.
3. Press `F4`.
4. Press `C-x e` to repeat the macro.

C. Rectangular editing

Rectangles are useful when editing columns of text.

Shortcut	Action
`C-x SPC`	Start rectangle selection (rectangle-mark-mode)
`C-x r k`	Kill rectangle
`C-x r y`	Yank rectangle
`C-x r d`	Delete rectangle
`C-x r t`	Replace rectangle contents with text
`C-x r N`	Insert line numbers in rectangle

D. Registers and bookmarks

Registers let you save positions or text for quick reuse.

Shortcut	Action
`C-x r SPC a`	Save current position in register a
`C-x r j a`	Jump to register a
`C-x r s a`	Save region in register a
`C-x r i a`	Insert register contents
`C-x r m`	Set bookmark
`C-x r b`	Jump to bookmark
`M-x bookmark-save`	Save bookmarks permanently

E. Help system

Emacs has an excellent built-in documentation system.

Shortcut	Action
`C-h t`	Interactive Emacs tutorial
`C-h k`	Describe a keyboard shortcut
`C-h f`	Describe a function
`C-h v`	Describe a variable
`C-h m`	Describe current major/minor modes
`C-h a`	Search commands by name

Shortcut	Action
`C-h i`	Open Emacs Info documentation
`M-x`	Execute any Emacs command by name

For example:

text

```
C-h k C-s
```

This tells you exactly what `C-s` does.

9. The most important shortcuts to memorize

If you don't want to memorize everything at once, start with these 25.

Emacs essential cheat sheet

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Movement
```

```
C-f
```

Forward character

```
C-b
```

Backward character

```
C-n
```

Next line

```
C-p
```

Previous line

```
C-a
```

Beginning of line

```
C-e
```

End of line

```
M-f
```

Forward word

```
M-b
```

Backward word

M-\<

Beginning of buffer

M-\>

End of buffer

```
Editing
```

Search and files

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Windows
```

Copy cheat sheet

10. Practice exercises

Try these inside Emacs to build muscle memory.

1. Movement: Open a file and navigate using only `C-f`, `C-b`, `C-n`, `C-p`, `M-f`, and `M-b`.
2. Editing: Delete a word, kill a line, undo, and paste it somewhere else.
3. Selection: Select a paragraph, copy it, and paste it elsewhere.
4. Search: Find a word and replace all its occurrences.
5. Windows: Open two files side by side and switch between them.
6. Buffers: Open three files and switch between their buffers.
7. Advanced: Record a keyboard macro that inserts a repeated code pattern.

Official learning resources

- [GNU Emacs Guided Tour](<https://www.gnu.org/software/emacs/tour/index.html?utm_source=chatgpt.com>) — excellent practical introduction.
- [GNU Emacs Manual](<https://www.gnu.org/software/emacs/manual/html_node/emacs/?utm_source=chatgpt.com>) — complete reference.
- [Emacs Docs](<https://emacsdocs.com/?utm_source=chatgpt.com>) — documentation organized by task.

My recommendation: Start with `C-f`, `C-b`, `C-n`, `C-p`, `C-a`, `C-e`, `M-f`, and `M-b`. These eight movement commands will become the foundation of your Emacs workflow. Also run `C-h t` inside Emacs to practice interactively.